

Eduardo Cintora (A17862193)

FIND A GENE PROJECT Part 1

[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.

Name: Neurofibromin

Accession: NP_001035957.1

Species: Homo Sapiens

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Blast Method: TBLASTN 2.17.0

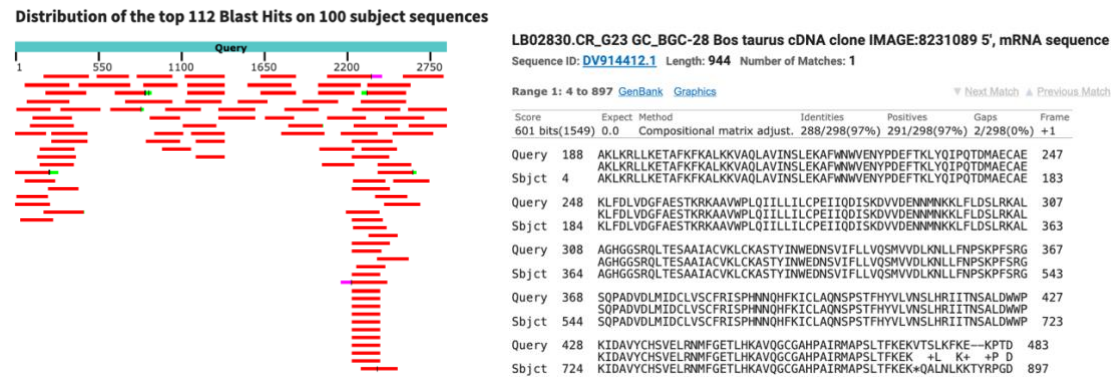
Database: expressed sequence tag (EST) database

Limits: No limits applied to any organism type

The screenshot shows the NCBI BLAST search interface. At the top, there are tabs for 'blastn', 'blastp', 'blastx', 'tblastn' (selected), and 'tblastx'. The 'tblastn' tab is active, and the text 'TBLASTN search translated' is visible in the top right corner. Below the tabs, the 'Enter Query Sequence' section contains a text box with the accession number 'NP_001035957.1'. To the right of this text box are 'Query subrange' fields for 'From' and 'To'. Below the text box is a 'Choose File' button and a 'No file chosen' status. The 'Job Title' section has a text box with the text 'NP_001035957-neurofibromin isoform 1 Homo...' and a 'Clear' button. Below the job title is a checkbox for 'Align two or more sequences'. The 'Choose Search Set' section has a 'Database' dropdown menu set to 'Expressed sequence tags (est)'. Below this are fields for 'Organism' (with a text box and an 'Add Organism' button), 'Exclude' (with checkboxes for 'Models (XM/XP)' and 'Uncultured/environmental sample sequences'), 'Limit to' (with a checkbox for 'Sequences from type material'), and 'Entrez® Query' (with a text box and a 'Create custom database' button). At the bottom, there is a 'BLAST' button and a checkbox for 'Show results in a new window'.

Chosen Match: Accession **DV914412.1**, a 944 base pair clone from *Bos taurus*.

Additional alignment details are inserted below.



Alignment Details:

>LB02830.CR_G23 GC_BGC-28 Bos taurus cDNA clone IMAGE:8231089 5', mRNA sequence

Sequence ID: DV914412.1 Length: 944

Range 1: 4 to 897

Score: 601 bits (1549), Expect: 0.0, Method: Compositional matrix adjust.,

Identities: 288/298 (97%), Positives: 291/298 (97%), Gaps: 2/298 (0%)

Query 188 AKLKRLLKETAFKFKALKKKVAQLAVINSLEKAFWNWVENYPDEFTKLYQIPQTDMAECAE 247
AKLKRLLKETAFKFKALKKKVAQLAVINSLEKAFWNWVENYPDEFTKLYQIPQTDMAECAE
Sbjct 4 AKLKRLLKETAFKFKALKKKVAQLAVINSLEKAFWNWVENYPDEFTKLYQIPQTDMAECAE 183

Query 248 KLFDLVDGFAESTKRKAAVWPLQIILLILCPEIIQDISKDVVDENNMNKKLFLDSLRLKAL 307
KLFDLVDGFAESTKRKAAVWPLQIILLILCPEIIQDISKDVVDENNMNKKLFLDSLRLKAL
Sbjct 184 KLFDLVDGFAESTKRKAAVWPLQIILLILCPEIIQDISKDVVDENNMNKKLFLDSLRLKAL 363

Query 308 AGHGGSRQLTESAAIACVKLCKASTYINWEDNSVIFLLVQSMVVDLKNLLFNPSKPFSSRG 367
AGHGGSRQLTESAAIACVKLCKASTYINWEDNSVIFLLVQSMVVDLKNLLFNPSKPFSSRG
Sbjct 364 AGHGGSRQLTESAAIACVKLCKASTYINWEDNSVIFLLVQSMVVDLKNLLFNPSKPFSSRG 543

Query 368 SQPADVDLMIDCLVSCFRISPHNNQHFKICLAQNSPSTFHYYLVNSLHRIITNSALDWWP 427

Eduardo Cintora (A17862193)

SQPADVDLMIDCLVSCFRISPHNNQHFKICLAQNSPSTFHYVLVNSLHRIITNSALDWWP

Sbjct 544 SQPADVDLMIDCLVSCFRISPHNNQHFKICLAQNSPSTFHYVLVNSLHRIITNSALDWWP 723

Query 428 KIDAVYCHSVELRNMFGETLHKAVQGCGAHPAIRMASLTFKEKVTSLKFKE--KPTD 483

KIDAVYCHSVELRNMFGETLHKAVQGCGAHPAIRMASLTFKEK +L K+ +P D

Sbjct 724 KIDAVYCHSVELRNMFGETLHKAVQGCGAHPAIRMASLTFKEK*QALNLKKTYRPGD 897

Alignment statistics: E-value of 0.0 with percent identity of 96.64% and a 10% query coverage.

[Q3] Gather information about this “novel” protein. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don’t forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don’t have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

Chosen Sequence:

This was obtained utilizing the FASTA format from BLAST results

```
>DV914412.1 LB02830.CR_G23 GC_BGC-28 Bos taurus cDNA clone IMAGE:8231089 5',  
mRNA sequence  
GATGCAAAATTAAAACGACTCCTGAAGGAAACAGCATTTTAAATTTAAAGCCCTAAAGAAAGTTGCACAGT  
TAGCAGTTATAAATAGCCTGGAAAAAGCATTGTTGGAAGTGGGTAGAAAATTATCCAGATGAATTTACAAA  
GCTGTACCAGATTCCACAGACTGATATGGCTGAGTGTGCTGAAAAACTGTTTGACTTAGTGGATGGTTTT  
GCTGAAAGCACCAAACGTAAAGCAGCAGTTTGGCCACTACAAATCATTCTGCTTATCTTGTGTCCAGAAA  
TAATCCAGGATATATCTAAGGATGTGGTTGATGAAAACAACATGAATAAGAAGTTGTTTCTGGACAGTCT  
GCGAAAAGCACTTGCCGGCCATGGAGGAAGCAGGCAGCTCACTGAAAGTGCTGCAATCGCCTGTGTCAA  
CTGTGTAAAGCAAGTACTTACATCAATTGGGAAGATAACTCTGTCATCTTTCTGCTGGTTCAGTCCATGG  
TGGTTGATCTTAAGAACCTGCTGTTTAATCCGAGTAAGCCATTCTCGAGGGGCAGTCAGCCGGCAGATGT  
GGACCTAATGATCGACTGCCTTGTCTCTTGCTTCCGGATAAGCCCTCACAATAACCAACACTTCAAGATC  
TGCCTGGCTCAGAATTTCCCTTCTACATTTCACTATGTGCTGGTAAATTCCTCCATCGAATCATCACC  
ATTCTGCATTGGATTGGTGGCCTAAGATTGATGCTGTATATTGCCACTCAGTTGAACTTCGAAATATGTT  
TGGTGAAACACTTCATAAAGCAGTCCAGGGTTGTGGAGCACACCCAGCAATACGAATGGCACCAGAGTCTT  
ACATTTAAAGAAAAGTAACAAGCCTTAAATTTAAAGAAAACCTACAGACCTGGAGACAAGAAGCTACAAG  
TATCTTCTCTTGTCCATGGTGAAACTAATTCATG
```

Protein sequence:

Eduardo Cintora (A17862193)

Achieved using the EMBOSS Transeq tool

```
>Bos Taurus cDNA clone
DAKLKRLKETAFAFKALKKKVAQLAVINSLEKAFWNWVENYPDEFTKLYQIPQTDMAECA
EKLFDLVDGFAESTKRKAAVWPLQIILLILCPEIIQDISKDVVDENNMNKKLFLDSLRLKA
LAGHGGSQRQLTESAAIACVKLCKASTYINWEDNSVIFLLVQSMVVDLKNLLFNPSKPFSSR
GSQPADVDLMIDCLVSCFRISPHNNQHFKICLAQNSPSTFHYVLVNSLHRIITNSALDWW
PKIDAVYCHSVELRNMFGETLHKAVQGCGAHPAIRMASLTFKEK*QALNLKKTYRPGDK
KLQVSSLVHGETNSX
```

Species: Bos Taurus

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria;
Laurasiatheria; Artiodactyla; Ruminantia; Pecora; Bovidae; Bovinae: Bos.

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, “novel” is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ? Clear

> Bos taurus cDNA clone
DAKLKRLKETAFAFKALKKKVAQLAVINSLEKAFWNWVENYPDEFTKLYQIP
QTDMAECAEKLFDLVDGFAESTKRKAAVWPLQIILLILCPEIIQDISKDVVDEN
NMNKKLFLDSLRLKALAGHGGSQRQLTESAAIACVKLCKASTYINWEDNSVIFL

Query subrange ?
From
To

Or, upload file

Choose File No file chosen ?

Job Title

Bos taurus cDNA clone

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database

Non-redundant protein sequences (nr) ?

Organism
Optional

Enter organism name or id—completions will be suggested ☐ exclude (
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. ?

Exclude
Optional

☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/en

Blastp search output hits:

☒ select all 100 sequences selected										GenPept	Graphics	Distance tree of results	Multiple alignment	MSA Viewer
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession					
☒	hypothetical protein G4228_003606 [Cervus hanglu yarkandensis]	Cervus hanglu y...	602	602	95%	0.0	96.64%	2503	KAF4012121.1					
☒	neurofibromin isoform X3 [Vulpes lagopus]	Vulpes lagopus	602	602	95%	0.0	96.64%	2834	XP_041582912.1					
☒	neurofibromin isoform X5 [Sagmatias obliquus]	Sagmatias obliq...	602	602	95%	0.0	96.64%	2688	XP_026986149.1					
☒	neurofibromin isoform X4 [Canis lupus dingo]	Canis lupus dingo	602	602	95%	0.0	96.64%	2840	XP_048970184.1					
☒	neurofibromin isoform X5 [Delphinus delphis]	Delphinus delphis	602	602	95%	0.0	96.64%	2688	XP_059852349.1					
☒	neurofibromin isoform X2 [Lipotes vexillifer]	Lipotes vexillifer	602	602	95%	0.0	96.64%	2820	XP_007452454.1					
☒	neurofibromin isoform X5 [Prionailurus viverrinus]	Prionailurus vive...	602	602	95%	0.0	96.64%	2792	XP_047687768.1					
☒	neurofibromin isoform 12-T12 [Hipposideros larvatus]	Hipposideros lar...	602	602	95%	0.0	96.64%	2838	KAM5215734.1					
☒	neurofibromin isoform X1 [Capricornis sumatraensis]	Capricornis sum...	602	602	95%	0.0	96.64%	2841	XP_068834314.1					
☒	neurofibromin isoform X6 [Ovis canadensis]	Ovis canadensis	602	602	95%	0.0	96.64%	2840	XP_069459009.1					
☒	neurofibromin isoform 3 [Leucopleurus acutus]	Leucopleurus ac...	602	602	95%	0.0	96.64%	2820	XP_082716647.1					
☒	neurofibromin isoform X6 [Symphalangus syndactylus]	Symphalangus s...	602	602	95%	0.0	96.64%	2706	XP_063484677.1					
☒	neurofibromin isoform X2 [Neophocaena asiaeorientalis asiaeorientalis]	Neophocaena a...	602	602	95%	0.0	96.64%	2840	XP_024592423.1					

Top Alignment Hit: hypothetical protein G4228_003606

Sequence ID: KAF4012121.1

The top result is for a protein from *Cervus hanglu yarkandensis* (Yarkand deer) species.

Statistics: E-value = 0.0 with a percent identity of 96.64% and query coverage of 95%.

[Download](#) [GenPept](#) [Graphics](#)

hypothetical protein G4228_003606 [Cervus hanglu yarkandensis]
Sequence ID: [KAF4012121.1](#) Length: 2503 Number of Matches: 1

Range 1: 188 to 483 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
602 bits(1553)	0.0	Compositional matrix adjust.	288/298(97%)	291/298(97%)	2/298(0%)
Query 2		AKLKRLK KETAFKFKALKKKVAQLAVINSLEKAFWNWVENYPDEFTKLYQIPQTDMAECAE			61
Sbjct 188		AKLKRLK KETAFKFKALKKKVAQLAVINSLEKAFWNWVENYPDEFTKLYQIPQTDMAECAE			247
Query 62		KLFDLV DVGFAESTKRKA AVWPLQI ILLILCPEIIQDISKDVVDENN MNKKLFLDSL RKAL			121
Sbjct 248		KLFDLV DVGFAESTKRKA AVWPLQI ILLILCPEIIQDISKDVVDENN MNKKLFLDSL RKAL			307
Query 122		AGHGGS RQLTESAAIACVKLCKASTYINWEDNSVIFLLVQSMVVDLKNLLFNPSKPF SRG			181
Sbjct 308		AGHGGS RQLTESAAIACVKLCKASTYINWEDNSVIFLLVQSMVVDLKNLLFNPSKPF SRG			367
Query 182		SQPADVD LMIDCLVSCFRISPHNNQHFKICLAQNSPSTFHYVLVNSLHRIITNSALDWWP			241
Sbjct 368		SQPADVD LMIDCLVSCFRISPHNNQHFKICLAQNSPSTFHYVLVNSLHRIITNSALDWWP			427
Query 242		KIDAVYCHSVELRNMFGETLHKAVQGC GAHPAIRM APSLTFKEK*QALNLKKT YRPGD			299
Sbjct 428		KIDAVYCHSVELRNMFGETLHKAVQGC GAHPAIRM APSLTFKEK +L K+ +P D			483